Statistics/Data analysis

```
      name:  SimulationBaltic
       log:  C:\Users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\ou
> tput.smcl
  log type:  smcl
 opened on:  26 Aug 2026, 19:31:59

1 .
2 . asdoc, text(\par \qc Financial Sustainability of the Care for the Disabled: A Simula
  > tion of Benefits for Older People in the Baltic States up to 2036 with data from SHA
  > RE and Eurostat) fs(12)  save(report.doc) replace
  (file report.doc not found)
  Click to Open File:  report.doc

3 . asdoc, text(\par \qc Hans Gevers - Junior Research Fellow at the Estonian Business S
  > chool https://orcid.org/0009-0001-0249-4142 hans.gevers@ebs.ee) fs(10)  save(report.
  > doc) append
  Click to Open File:  report.doc

4 .
5 . *>>>PREPROCESSING
6 .
7 . foreach num of numlist 8(1)9{
    2.          clear all
    3.          cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data
  > 7\sharew`num'_rel9-0-0_ALL_datasets_stata\"
    4.          use "sharew`num'_rel9-0-0_gv_imputations.dta"
    5.          keep if implicat==1
    6.          merge 1:1 mergeid using "sharew`num'_rel9-0-0_gv_health.dta"
    7.          isid mergeid
    8.          drop _merge
    9.          merge 1:1 mergeid using "sharew`num'_rel9-0-0_gv_weights.dta"
   10.          isid mergeid
   11.          drop _merge
   12.
8 .         if `num'==8{
   13.                  gen Qyear=2020
   14.          }
   15.          if `num'==9{
   16.                  gen Qyear=2022
   17.          }
   18.
9 .         keep if country==57 | country==48 | country==35
   19.
10.         save "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7
  > \W`num'_final.dta", replace
   20. }
  C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\sharew8_rel9-0-
  > 0_ALL_datasets_stata
  (214,780 observations deleted)
  (label orienti already defined)
  (label eurod already defined)
  (label numeracy already defined)
  (label gali already defined)
  (label language already defined)
  (label country already defined)

      Result                      Number of obs
      -----------------------------------------
      Not matched                             0
      Matched                            53,695  (_merge==3)
      -----------------------------------------

  (label country already defined)
  (label language already defined)
```

```
    Result                      Number of obs
    -----------------------------------------
    Not matched                             0
    Matched                            53,695  (_merge==3)
    -----------------------------------------

(47,632 observations deleted)
file C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data
    7\W8_final.dta saved
C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\sharew9_rel9-0-
> 0_ALL_datasets_stata
(277,788 observations deleted)
(label orienti already defined)
(label eurod already defined)
(label numeracy already defined)
(label gali already defined)
(label language already defined)
(label country already defined)

    Result                      Number of obs
    -----------------------------------------
    Not matched                             0
    Matched                            69,447  (_merge==3)
    -----------------------------------------

(label country already defined)
(label language already defined)

    Result                      Number of obs
    -----------------------------------------
    Not matched                             0
    Matched                            69,447  (_merge==3)
    -----------------------------------------

(61,831 observations deleted)
file C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data
    7\W9_final.dta saved

11.
12. clear all

13. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\"
C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7

14. use "W8_final.dta"

15. merge 1:1 mergeid using "weightslong.dta", keep(match)
(label country already defined)

    Result                      Number of obs
    -----------------------------------------
    Not matched                             0
    Matched                             5,101  (_merge==3)
    -----------------------------------------

16. drop _merge

17. save "W8_final.dta", replace
file W8_final.dta saved

18.
19. clear all
```

```
20. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\"
  C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7

21. use "W9_final.dta"

22. merge 1:1 mergeid using "weightslong.dta", keep(match)
  (label country already defined)

      Result                           Number of obs
      -----------------------------------------
      Not matched                                  0
      Matched                                  4,921  (_merge==3)
      -----------------------------------------

23. drop _merge

24. save "W9_final.dta", replace
  file W9_final.dta saved

25.
26. clear all

27. cd "C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\"
  C:\users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7

28. use "W8_final.dta"

29. append using "W9_final.dta"
  (variable psu was str10, now str11 to accommodate using data's values)
  (variable ssu was str6, now str13 to accommodate using data's values)
  (label pstat already defined)
  (label language already defined)
  (label country already defined)
  (label Hrespt already defined)
  (label pcskill already defined)
  (label IMPflag already defined)
  (label orienti already defined)
  (label bfi10 already defined)
  (label thospital already defined)
  (label satisfied already defined)
  (label often already defined)
  (label yedu already defined)
  (label fdistress already defined)
  (label writing already defined)
  (label reading already defined)
  (label hearing already defined)
  (label eyesight already defined)
  (label mstat already defined)
  (label sphus already defined)
  (label noyes already defined)
  (label hnrsc already defined)
  (label undersq already defined)
  (label clarif already defined)
  (label willans already defined)
  (label otrf already defined)
  (label empstat already defined)
  (label cjs already defined)
  (label isced already defined)
  (label gender already defined)
  (label num already defined)
  (label gali already defined)
  (label meat already defined)
  (label legeggs already defined)
  (label fruit already defined)
  (label numeracy already defined)
  (label numeracy2 already defined)
  (label eurod already defined)
  (label dairyp already defined)
  (label weekday already defined)
  (label specday already defined)
  (label dkrf already defined)
  (label dummi already defined)
```

```
  (label zero_one already defined)
  (label loneli already defined)
  (label chronic2 already defined)
  (label sphus2 already defined)
  (label bmi2 already defined)
  (label mobilit2 already defined)
  (label mobilit3 already defined)
  (label adl2 already defined)
  (label iadl2 already defined)
  (label phactiv already defined)

30.
31. save "dataset.dta", replace
  file dataset.dta saved

32.
33. clear all

34.
35. use "dataset.dta"

36.
37. *>>>PREPARATION
38.
39. generate income = thinc-ypen3

40.
41. egen id= group(mergeid)

42.
43. keep ypen3 gender age country income gali id Qyear

44.
45. replace income=0 if income<0
  (44 real changes made)

46.
47. drop if ypen3>7000
  (7 observations deleted)

48.
49. *additional age categorization to match Eurostat categories
50. generate ageD=0

51. replace ageD=1 if age>=55 & age<=64
  (2,793 real changes made)

52. replace ageD=2 if age>=65 & age<=74
  (3,142 real changes made)

53. replace ageD=3 if age>=75 & age<=84
  (2,554 real changes made)

54. replace ageD=4 if age>=85
  (876 real changes made)

55. label define ageDl 0 "Less than 55 years old" 1 "Between 55 and 64 years old" 2 "Bet
  > ween 65 and 74 years old" 3 "Between 75 and 84 years old" 4 "Older than 85 years"

56. label values ageD ageDl
```

```
57.
58. generate ageE=0

59. replace ageE=1 if age>=55 & age<=64
  (2,793 real changes made)

60. replace ageE=2 if age>=65
  (6,572 real changes made)

61. label define agel 0 "Less than 55 years old" 1 "Between 55 and 64 years old" 2 "Olde
  > r than 64 years"

62. label values ageE agel

63.
64. *>>>DESCRIPTIVES
65.
66. tabulate country
```

Country identifier	Freq.	Percent	Cum.
Estonia	5,573	55.65	55.65
Lithuania	2,429	24.25	79.90
Latvia	2,013	20.10	100.00
Total	10,015	100.00	

```
67.
68. asdoc codebook ypen3 gender age country income gali id Qyear, compact save(report.do
  > c) append
```

Variable	Obs	Unique	Mean	Min	Max	Label
ypen3	10015	382	232.8375	0	6816	Disability pensions/benefits
gender	10015	2	1.632052	1	2	Gender
age	10015	65	69.76246	34	102	Age
country	10015	3	42.57494	35	57	Country identifier
income	10015	4468	10071.76	0	170698	
gali	10015	2	.5899151	0	1	Limitation with activities
id	10015	5101	2556.266	1	5101	group(mergeid)
Qyear	10015	2	2020.982	2020	2022	

Click to Open File: report.doc

```
69.
70. asdoc codebook, save(report.doc) append
```

country **Country identifier**

```
                 Type: Numeric (byte)
                Label: country

                Range: [35,57]                      Units: 1
        Unique values: 3                        Missing .: 0/10,015

           Tabulation: Freq.   Numeric  Label
                       5,573        35  Estonia
                       2,429        48  Lithuania
                       2,013        57  Latvia
```

ypen3 **Disability pensions/benefits**

```
           Type: Numeric (double)

          Range: [0,6816]                    Units: 1.000e-07
  Unique values: 382                     Missing .: 0/10,015

           Mean: 232.838
      Std. dev.: 885.214

    Percentiles:     10%       25%       50%       75%       90%
                       0         0         0         0       276
```

---

gender **Gender**

```
           Type: Numeric (byte)
          Label: gender

          Range: [1,2]                       Units: 1
  Unique values: 2                       Missing .: 0/10,015

     Tabulation: Freq.   Numeric  Label
                 3,685         1  Male
                 6,330         2  Female
```

---

age **Age**

```
           Type: Numeric (int)
          Label: num, but 65 nonmissing values are not labeled

          Range: [34,102]                    Units: 1
  Unique values: 65                      Missing .: 0/10,015

       Examples: 60
                 66
                 73
                 80
```

---

gali **Limitation with activities**

```
           Type: Numeric (byte)
          Label: gali

          Range: [0,1]                       Units: 1
  Unique values: 2                       Missing .: 0/10,015

     Tabulation: Freq.   Numeric  Label
                 4,107         0  Not limited
                 5,908         1  Limited
```

Qyear **(unlabeled)**

```
           Type: Numeric (float)

          Range: [2020,2022]                 Units: 1
  Unique values: 2                       Missing .: 0/10,015

     Tabulation: Freq.  Value
                 5,100  2020
                 4,915  2022
```

---

income **(unlabeled)**

Type: Numeric (**float**)

Range: [**0,170697.95**] Units: **1.000e-20**
Unique values: **4,468** Missing .: **0**/**10,015**

Mean: **10071.8**
Std. dev.: **8975.29**

Percentiles:	10%	25%	50%	75%	90%
	1968	**4900**	**7750**	**12860**	**19971.8**

---

id **group(mergeid)**

Type: Numeric (**float**)

Range: [**1,5101**] Units: **1**
Unique values: **5,101** Missing .: **0**/**10,015**

Mean: **2556.27**
Std. dev.: **1474.24**

Percentiles:	10%	25%	50%	75%	90%
	512	**1278**	**2559**	**3835**	**4595**

---

ageD **(unlabeled)**

Type: Numeric (**float**)
Label: **ageDl**

Range: [**0,4**] Units: **1**
Unique values: **5** Missing .: **0**/**10,015**

Tabulation: Freq.	Numeric	Label
650	**0**	Less than 55 years old
2,793	**1**	Between 55 and 64 years old
3,142	**2**	Between 65 and 74 years old
2,554	**3**	Between 75 and 84 years old
876	**4**	Older than 85 years

---

ageE **(unlabeled)**

Type: Numeric (**float**)
Label: **agel**

Range: [**0,2**] Units: **1**
Unique values: **3** Missing .: **0**/**10,015**

Tabulation: Freq.	Numeric	Label
650	**0**	Less than 55 years old
2,793	**1**	Between 55 and 64 years old
6,572	**2**	Older than 64 years

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71.

```
72. hist ypen3 if ypen3>0, scheme(s2color) xlabel(0(1000)7000,angle(0) labsize(small) gr
  > id) ylabel(0.000(0.0002)0.001,angle(0) labsize(small) grid format(%9.0gc))
  (bin=30, start=40, width=225.86667)

73. graph export benefits.svg, replace height(2400)
  file benefits.svg saved as SVG format

74. vioplot age if Qyear==2020, over(gender) over(country) scheme(s2color) ytitle("Numbe
  > r of respondents", margin(small)) ylabel(30(10)100,angle(0) labsize(small) grid)

75. graph export sample.svg, replace height(2400)
  file sample.svg saved as SVG format

76. vioplot age if Qyear==2020 & ypen3>0, over(gender) over(country) scheme(s2color) yti
  > tle("Number of respondents", margin(small)) ylabel(30(10)100,angle(0) labsize(small)
  >  grid)

77. graph export sampleDis.svg, replace height(2400)
  file sampleDis.svg saved as SVG format

78.
79. asdoc spearman ypen3 gender age country income gali id Qyear, save(report.doc) appen
  > d

  Number of observations = 10,015

               |    ypen3   gender      age  country   income     gali       id    Qyear
  -------------+------------------------------------------------------------------------
         ypen3 |   1.0000
        gender |  -0.0184   1.0000
           age |  -0.1637   0.0735   1.0000
       country |  -0.0932  -0.0113  -0.0870   1.0000
        income |  -0.1245  -0.1481  -0.0799  -0.3425   1.0000
          gali |   0.2259   0.0210   0.2934  -0.0195  -0.1472   1.0000
            id |  -0.0918  -0.0058  -0.0739   0.8974  -0.2985  -0.0242   1.0000
         Qyear |  -0.0373   0.0143   0.0877   0.0042   0.0780   0.0266   0.0037   1.0000
  >
  Click to Open File:  report.doc

80.
81. foreach ageC in ageD ageE{
  2.         foreach num of numlist 35 48 57{
  3.                         foreach num2 of numlist 1 2{
  4.                                 foreach num3 of numlist 0 1{
  5.                                         asdoc, text(\par for `ageC': Country `num
  > ' , gender `num2' , limited `num3')
  6.                                         asdoc tabstat income if country==`num' &
  > gender==`num2' & gali==`num3', by(`ageC') stat(mean), save(report.doc) append
  7.                                 }
  8.                         }
  9.         }
 10. }
  (File report.doc already exists, option append was assumed)
  Click to Open File:  report.doc
  ageD

               |    income
  -------------+----------
  Between_55~d |  20571.48
  Between_65~d |  17941.57
  Between_75~d |  16346.79
  Less_than_~d |  12735.02
  Older_than~s |  11303.48
  Click to Open File:  report.doc
  (File report.doc already exists, option append was assumed)
  Click to Open File:  report.doc
  ageD
```

	income
Between_55~d	**12357.32**
Between_65~d	**13166.79**
Between_75~d	**13350.15**
Less_than_~d	**11962.99**
Older_than~s	**10109.23**

Click to Open File: report.doc
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ageD

	income
Between_55~d	**20010.27**
Between_65~d	**16784.18**
Between_75~d	**13343.93**
Less_than_~d	**9293.497**
Older_than~s	**7288.414**

Click to Open File: report.doc
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ageD

	income
Between_55~d	**13631.43**
Between_65~d	**10804.44**
Between_75~d	**10525.3**
Less_than_~d	**8400.997**
Older_than~s	**7017.735**

Click to Open File: report.doc
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ageD

	income
Between_55~d	**7846.787**
Between_65~d	**11125**
Between_75~d	**9710.709**
Less_than_~d	**9511.654**
Older_than~s	**7384.786**

Click to Open File: report.doc
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ageD

	income
Between_55~d	**9433.906**
Between_65~d	**7559.58**
Between_75~d	**7553.142**
Less_than_~d	**7746.841**
Older_than~s	**6844.66**

Click to Open File: report.doc
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ageD

	income
Between_55~d	**7358.451**
Between_65~d	**9387.978**
Between_75~d	**8545.833**
Less_than_~d	**6252.98**
Older_than~s	**5745.49**

Click to Open File: report.doc
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Click to Open File: report.doc
ageD

	income
Between_55~d	**9299.859**
Between_65~d	**7512.068**
Between_75~d	**6602.537**
Less_than_~d	**6118.597**
Older_than~s	**4264.137**

Click to Open File: report.doc
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Click to Open File: report.doc
ageD

	income
Between_55~d	**3934.908**
Between_65~d	**7333.089**
Between_75~d	**9284.544**
Less_than_~d	**9234.106**
Older_than~s	**7375.984**

Click to Open File: report.doc
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ageD

	income
Between_55~d	**5057.722**
Between_65~d	**5015.384**
Between_75~d	**8295.423**
Less_than_~d	**7732.316**
Older_than~s	**7267.691**

Click to Open File: report.doc
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ageD

	income
Between_55~d	**7083.539**
Between_65~d	**6286.68**
Between_75~d	**8461.983**
Less_than_~d	**6394.999**
Older_than~s	**5576.932**

Click to Open File: report.doc
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ageD

	income
Between_55~d	**5664.375**
Between_65~d	**5343.996**
Between_75~d	**6306.117**
Less_than_~d	**5626.213**
Older_than~s	**4190.057**

Click to Open File: report.doc
(File report.doc already exists, option **append** was assumed)
Click to Open File: report.doc
ageE

	income
Between_55~d	**20571.48**
Less_than_~d	**17941.57**
Older_than~s	**14906.06**

Click to Open File: report.doc
(File report.doc already exists, option **append** was assumed)
Click to Open File: report.doc
ageE

	income
Between_55~d	**12357.32**
Less_than_~d	**13166.79**
Older_than~s	**12353.23**

Click to Open File: report.doc
(File report.doc already exists, option **append** was assumed)
Click to Open File: report.doc
ageE

	income
Between_55~d	**20010.27**
Less_than_~d	**16784.18**
Older_than~s	**11453.5**

Click to Open File: report.doc
(File report.doc already exists, option **append** was assumed)
Click to Open File: report.doc
ageE

	income
Between_55~d	**13631.43**
Less_than_~d	**10804.44**
Older_than~s	**8946.796**

Click to Open File: report.doc
(File report.doc already exists, option **append** was assumed)
Click to Open File: report.doc
ageE

	income
Between_55~d	**7846.787**
Less_than_~d	**11125**
Older_than~s	**9552.095**

Click to Open File: report.doc
(File report.doc already exists, option **append** was assumed)
Click to Open File: report.doc
ageE

	income
Between_55~d	**9433.906**
Less_than_~d	**7559.58**
Older_than~s	**7541.938**

Click to Open File: report.doc
(File report.doc already exists, option **append** was assumed)
Click to Open File: report.doc
ageE

	income
Between_55~d	**7358.451**
Less_than_~d	**9387.978**
Older_than~s	**7588.958**

Click to Open File: report.doc
(File report.doc already exists, option **append** was assumed)
Click to Open File: report.doc
ageE

	income
Between_55~d	**9299.859**
Less_than_~d	**7512.068**
Older_than~s	**5866.109**

Click to Open File: report.doc
(File report.doc already exists, option **append** was assumed)
Click to Open File: report.doc
ageE

```
                 |     income
-----------------+-----------
Between_55~d     |   3934.908
Less_than_~d     |   7333.089
Older_than~s     |   9193.949
Click to Open File:  report.doc
(File report.doc already exists, option append was assumed)
Click to Open File:  report.doc
ageE

                 |     income
-----------------+-----------
Between_55~d     |   5057.722
Less_than_~d     |   5015.384
Older_than~s     |    7950.08
Click to Open File:  report.doc
(File report.doc already exists, option append was assumed)
Click to Open File:  report.doc
ageE

                 |     income
-----------------+-----------
Between_55~d     |   7083.539
Less_than_~d     |    6286.68
Older_than~s     |    7731.15
Click to Open File:  report.doc
(File report.doc already exists, option append was assumed)
Click to Open File:  report.doc
ageE

                 |     income
-----------------+-----------
Between_55~d     |   5664.375
Less_than_~d     |   5343.996
Older_than~s     |   5703.856
Click to Open File:  report.doc

82.
83. *>>>ANALYSIS
84.
85. xtset id Qyear

  Panel variable: id (unbalanced)
   Time variable: Qyear, 2020 to 2022, but with gaps
           Delta: 1 unit

86.
87. *for disability
88. xttobit ypen3 i.gender i.ageD i.gali income i.country, ll(0) iterate(25) tobit

  Fitting comparison model:

  Fitting constant-only model:

  Iteration 0:  Log likelihood = -21562.521
  Iteration 1:  Log likelihood = -13222.906
  Iteration 2:  Log likelihood =  -13170.91
  Iteration 3:  Log likelihood = -13170.391
  Iteration 4:  Log likelihood = -13170.391

  Fitting full model:

  Iteration 0:  Log likelihood =   -21003.9
  Iteration 1:  Log likelihood = -12648.454
  Iteration 2:  Log likelihood =  -12233.36
  Iteration 3:  Log likelihood = -12215.992
  Iteration 4:  Log likelihood = -12215.928
  Iteration 5:  Log likelihood = -12215.928

  Obtaining starting values for full model:
```

```
Iteration 0:  Log likelihood = -80432.953
Iteration 1:  Log likelihood =  -80374.31
Iteration 2:  Log likelihood = -80373.876
Iteration 3:  Log likelihood = -80373.876

Fitting full model:

Iteration 0:  Log likelihood = -19495.179
Iteration 1:  Log likelihood = -14709.844  (not concave)
Iteration 2:  Log likelihood = -12911.337
Iteration 3:  Log likelihood = -12042.141
Iteration 4:  Log likelihood = -11985.302
Iteration 5:  Log likelihood = -11984.256
Iteration 6:  Log likelihood = -11984.255
Iteration 7:  Log likelihood = -11984.255

Random-effects tobit regression                     Number of obs     = 10,015
                                                           Uncensored =  1,120
Limits: Lower =    0                                    Left-censored =  8,895
        Upper = +inf                                   Right-censored =      0

Group variable: id                                  Number of groups  =  5,101
Random effects u_i ~ Gaussian                       Obs per group:
                                                                  min =      1
                                                                  avg =    2.0
                                                                  max =      2

Integration method: mvaghermite                     Integration pts.  =     12

                                                    Wald chi2(9)      = 810.68
Log likelihood = -11984.255                         Prob > chi2       = 0.0000

-----------------------------------------------------------------------------------------
  ---------
                       ypen3 | Coefficient  Std. err.      z    P>|z|     [95% conf. i
> nterval]
-----------------------------+-----------------------------------------------------------
  ---------
                      gender |
                     Female  |   -287.643   135.2631    -2.13   0.033    -552.7537   -
> 22.53216
                             |
                        ageD |
Between 55 and 64 years old  |   674.5725    220.543     3.06   0.002     242.3162
> 1106.829
Between 65 and 74 years old  |  -2665.123   249.0781   -10.70   0.000    -3153.307   -
> 2176.938
Between 75 and 84 years old  |  -3295.505   260.9746   -12.63   0.000    -3807.006   -
> 2784.004
        Older than 85 years  |  -3137.637   306.8433   -10.23   0.000    -3739.039   -
> 2536.236
                             |
                        gali |
                    Limited  |   3067.465   164.0657    18.70   0.000     2745.903
> 3389.028
                      income |  -.0813264   .0080558   -10.10   0.000    -.0971155   -
> .0655373
                             |
                     country |
                  Lithuania  |   -1361.48   169.4983    -8.03   0.000     -1693.69   -
> 1029.269
                     Latvia  |  -2580.163   208.8472   -12.35   0.000    -2989.496
> -2170.83
                             |
                       _cons |  -2954.945   287.3196   -10.28   0.000    -3518.082   -
> 2391.809
-----------------------------+-----------------------------------------------------------
  ---------
                    /sigma_u |   2498.727   91.03549    27.45   0.000     2320.301
> 2677.153
                    /sigma_e |   1874.845    58.8259    31.87   0.000     1759.548
> 1990.142
```

```
─────────────────────────────┼──────────────────────────────────────────────────
  ────────
                         rho │   .6398034   .0217323                      .5964239
> .6814528
─────────────────────────────┴──────────────────────────────────────────────────
  ────────
LR test of sigma_u=0: chibar2(01) = 463.35           Prob >= chibar2 = 0.000


89. outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
  > title(DisabilityTobit) replace
  results.xls
  dir : seeout

90.
91. xtpoisson ypen3 i.gender i.ageD i.gali income i.country if ypen3>0, re vce(robust)
  note: you are responsible for interpretation of non-count dep. variable

  Fitting Poisson model:

  Iteration 0:  Log pseudolikelihood = -510758.37
  Iteration 1:  Log pseudolikelihood = -509695.99
  Iteration 2:  Log pseudolikelihood = -509693.04
  Iteration 3:  Log pseudolikelihood = -509693.04

  Fitting full model:

  Iteration 0:  Log pseudolikelihood = -90029.433
  Iteration 1:  Log pseudolikelihood = -81819.234
  Iteration 2:  Log pseudolikelihood =  -81485.43
  Iteration 3:  Log pseudolikelihood = -81464.412
  Iteration 4:  Log pseudolikelihood = -81464.282
  Iteration 5:  Log pseudolikelihood = -81464.282

  Random-effects Poisson regression                  Number of obs    =    1,120
  Group variable: id                                 Number of groups =      769

  Random effects u_i ~ Gamma                         Obs per group:
                                                                  min =        1
                                                                  avg =      1.5
                                                                  max =        2

                                                     Wald chi2(9)     = 73075.05
  Log pseudolikelihood = -81464.282                  Prob > chi2      =   0.0000

                                                       (Std. err. adjusted for clusterin
  > g on id)
─────────────────────────────┬──────────────────────────────────────────────────
  ────────
                             │               Robust
                       ypen3 │ Coefficient  std. err.      z    P>|z|     [95% conf. i
  > nterval]
─────────────────────────────┼──────────────────────────────────────────────────
  ────────
                      gender │
                     Female  │  -.1144357   .0626614    -1.83   0.068    -.2372498
  > .0083784
                             │
                        ageD │
  Between 55 and 64 years old │   .3505128   .1000438     3.50   0.000     .1544306
  > .5465951
  Between 65 and 74 years old │  -.0381719   .2200633    -0.17   0.862     -.469488
  > .3931442
  Between 75 and 84 years old │  -.2965865   .2460516    -1.21   0.228    -.7788388
  > .1856659
          Older than 85 years │  -.1641732   .2578095    -0.64   0.524    -.6694706
  > .3411241
                             │
                        gali │
                    Limited  │   .0414885   .0798963     0.52   0.604    -.1151052
  > .1980823
                      income │  -3.51e-06   4.86e-06    -0.72   0.470     -.000013
```

```
> 6.01e-06
                         |
                 country |
               Lithuania |   .3884312   .0765913     5.07   0.000     .238315
> .5385474
                  Latvia |   .1006352   .0846788     1.19   0.235   -.0653322
> .2666026
                         |
                   _cons |   7.333101   .1339558    54.74   0.000    7.070553
>  7.59565
-------------------------+----------------------------------------------------------------
  --------
                /lnalpha |  -.2402887   .0997321                    -.4357599   -
> .0448174
-------------------------+----------------------------------------------------------------
  --------
                   alpha |   .7864008   .0784294                      .646773
> .9561721
-------------------------+----------------------------------------------------------------
  --------
LR test of alpha=0: chibar2(01) = 8.6e+05             Prob >= chibar2 = 0.000

92. outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
  > title(DisabilityPoisson) append
  results.xls
  dir : seeout

93. outreg2 using resultsNoStar.xls, excel dec(10) ctitle(DisabilityPoisson) noaster nos
  > e replace
  resultsNoStar.xls
  dir : seeout

94.
95. *for income
96. xttobit ypen3 i.gender i.ageE i.gali income i.country, ll(0) iterate(25) tobit

  Fitting comparison model:

  Fitting constant-only model:

  Iteration 0:  Log likelihood = -21562.521
  Iteration 1:  Log likelihood = -13222.906
  Iteration 2:  Log likelihood =  -13170.91
  Iteration 3:  Log likelihood = -13170.391
  Iteration 4:  Log likelihood = -13170.391

  Fitting full model:

  Iteration 0:  Log likelihood = -21016.972
  Iteration 1:  Log likelihood = -12656.011
  Iteration 2:  Log likelihood = -12239.864
  Iteration 3:  Log likelihood = -12222.092
  Iteration 4:  Log likelihood =  -12222.03
  Iteration 5:  Log likelihood =  -12222.03

  Obtaining starting values for full model:

  Iteration 0:  Log likelihood = -80443.279
  Iteration 1:  Log likelihood = -80388.043
  Iteration 2:  Log likelihood = -80387.672
  Iteration 3:  Log likelihood = -80387.672

  Fitting full model:
```

```
Iteration 0:  Log likelihood = -19496.048
Iteration 1:  Log likelihood =   -14692.5  (not concave)
Iteration 2:  Log likelihood = -13545.246
Iteration 3:  Log likelihood = -12045.587
Iteration 4:  Log likelihood = -11999.188
Iteration 5:  Log likelihood = -11990.142
Iteration 6:  Log likelihood = -11990.072
Iteration 7:  Log likelihood = -11990.072
Iteration 8:  Log likelihood = -11990.072

Random-effects tobit regression                     Number of obs     = 10,015
                                                           Uncensored =  1,120
Limits: Lower =    0                                    Left-censored =  8,895
        Upper = +inf                                   Right-censored =      0

Group variable: id                                  Number of groups  =  5,101
Random effects u_i ~ Gaussian                       Obs per group:
                                                                  min =      1
                                                                  avg =    2.0
                                                                  max =      2

Integration method: mvaghermite                     Integration pts.  =     12

                                                    Wald chi2(7)      = 797.62
Log likelihood = -11990.072                         Prob > chi2       = 0.0000

------------------------------------------------------------------------------
  ---------
                       ypen3 | Coefficient  Std. err.      z    P>|z|     [95% conf. i
> nterval]
-----------------------------+------------------------------------------------
  ---------
                      gender |
                     Female  |   -309.286   135.2379    -2.29   0.022    -574.3475   -
> 44.22447
                             |
                        ageE |
Between 55 and 64 years old  |   660.9327   220.8577     2.99   0.003     228.0596
> 1093.806
        Older than 64 years  |  -2970.574   235.0225   -12.64   0.000    -3431.209   -
> 2509.938
                             |
                        gali |
                    Limited  |   3047.965   164.3756    18.54   0.000     2725.795
> 3370.135
                      income |  -.0801059   .0080646    -9.93   0.000    -.0959123   -
> .0642995
                             |
                     country |
                  Lithuania  |  -1350.161   169.5277    -7.96   0.000     -1682.43   -
> 1017.893
                     Latvia  |  -2548.195   208.4733   -12.22   0.000    -2956.795   -
> 2139.595
                             |
                       _cons |  -2958.879   287.9623   -10.28   0.000    -3523.274   -
> 2394.483
-----------------------------+------------------------------------------------
  ---------
                    /sigma_u |   2504.145    91.0116    27.51   0.000     2325.766
> 2682.524
                    /sigma_e |   1880.215    59.0092    31.86   0.000      1764.56
> 1995.871
-----------------------------+------------------------------------------------
  ---------
                         rho |   .6394832   .0216781                      .5962159
> .6810337
------------------------------------------------------------------------------
  ---------
LR test of sigma_u=0: chibar2(01) = 463.92         Prob >= chibar2 = 0.000
```

```
97. outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
  > title(IncomeTobit) append
  results.xls
  dir : seeout

98.
99. xtpoisson ypen3 i.gender i.ageE i.gali income i.country if ypen3>0, re vce(robust)
  note: you are responsible for interpretation of non-count dep. variable

  Fitting Poisson model:

  Iteration 0:  Log pseudolikelihood = -535702.37
  Iteration 1:  Log pseudolikelihood = -535546.08
  Iteration 2:  Log pseudolikelihood = -535546.04

  Fitting full model:

  Iteration 0:  Log pseudolikelihood = -95899.544
  Iteration 1:  Log pseudolikelihood = -82803.035
  Iteration 2:  Log pseudolikelihood = -81530.966
  Iteration 3:  Log pseudolikelihood = -81502.661
  Iteration 4:  Log pseudolikelihood = -81502.309
  Iteration 5:  Log pseudolikelihood = -81502.309

  Random-effects Poisson regression                  Number of obs    =     1,120
  Group variable: id                                 Number of groups =       769

  Random effects u_i ~ Gamma                         Obs per group:
                                                                  min =         1
                                                                  avg =       1.5
                                                                  max =         2

                                                     Wald chi2(7)     =  74617.49
  Log pseudolikelihood = -81502.309                  Prob > chi2      =    0.0000

                                                        (Std. err. adjusted for clusterin
  > g on id)
  ───────────────────────────────┬──────────────────────────────────────────────────────
    ────────
                                 │               Robust
                           ypen3 │ Coefficient  std. err.      z    P>|z|     [95% conf. i
  > nterval]
  ───────────────────────────────┼──────────────────────────────────────────────────────
    ────────
                                 │
                          gender │
                          Female │  -.1189755   .0637245    -1.87   0.062    -.2438732
  > .0059222
                                 │
                            ageE │
  Between 55 and 64 years old    │   .3505488   .1000543     3.50   0.000     .1544461
  > .5466516
          Older than 64 years    │  -.0382013   .2202289    -0.17   0.862     -.469842
  > .3934394
                                 │
                            gali │
                         Limited │   .0414862   .0799038     0.52   0.604    -.1151224
  > .1980949
                          income │  -3.51e-06   4.86e-06    -0.72   0.471     -.000013
  > 6.02e-06
                                 │
                         country │
                       Lithuania │   .3722408   .0752281     4.95   0.000     .2247965
  >  .519685
                          Latvia │     .11164   .0846705     1.32   0.187     -.054311
  > .2775911
                                 │
                           _cons │   7.318747   .1334633    54.84   0.000     7.057164
  >  7.58033
  ───────────────────────────────┼──────────────────────────────────────────────────────
    ────────
                        /lnalpha │   -.176007   .0927892                     -.3578705
  > .0058565
```

```
──────────────────────────────┬───────────────────────────────────────────────────────────
  ────────
                        alpha │   .8386121   .0778141                      .6991636
> 1.005874
──────────────────────────────┴───────────────────────────────────────────────────────────
  ────────
LR test of alpha=0: chibar2(01) = 9.1e+05              Prob >= chibar2 = 0.000

100 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
  > title(IncomePoisson) append
  results.xls
  dir : seeout

101 outreg2 using resultsNoStar.xls, excel dec(10) ctitle(IncomePoisson) noaster nose ap
  > pend
  resultsNoStar.xls
  dir : seeout

102
103 log close SimulationBaltic
        name:  SimulationBaltic
         log:  C:\Users\hansg\Documents\bestanden\PhD ebs\JRF work Stata\SHARE data 7\ou
  > tput.smcl
    log type:  smcl
   closed on:  26 Aug 2026, 19:33:14
──────────────────────────────────────────────────────────────────────────────────────────
```